

Bio-Inspired Muscle Control of Quadruped Locomotion Using Central Pattern Generators

Motivation

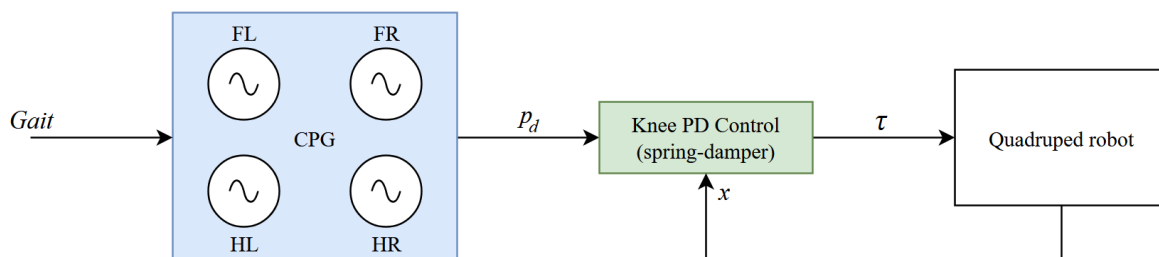
Traditional control approaches for legged robots typically rely on centralized methods such as finite state machine-based gait control or model-based gait control, which depend on accurate full-body dynamic models. While these methods can be effective, the complexity and sensitivity to modeling errors makes them less desirable. In nature, it is observed that vertebrates use Central Pattern Generators (CPGs) for coordinating movement. CPGs are small neural networks which produce rhythmic behavior with minimal sensory input. Motivated by the advantages observed in biological locomotion, this project explores a simple bio-inspired control architecture, without relying on a complex dynamic model.

Objective

This project aims to develop a bio-inspired locomotion control system for a quadruped robot, capable of producing multiple gait patterns.

Methodology

The control architecture combines a CPG, for the actively driven hip joints, with a muscle-inspired spring-damper PD controller for the passive knee joints. The CPG is used to generate coordinated rhythmic signals that define and modulate different locomotion gaits, meanwhile the PD controller emulates muscle-like compliance by allowing the knee joint to passively respond to the motion generated at the hip joint, as illustrated in the control diagram below. To evaluate the designed control architecture, its performance will be compared against Virtual Model Control (VMC), a control method that creates virtual forces to adjust joint torques.



Control Diagram, illustrating the control flow of the project

Timeline	
Task	Start date
Research and detailed planning	02/02-26 (Week 1)
Creating the CPG	09/02-26 (Week 2)
Creating the muscle (spring-damper) PD controller	02/03-26 (Week 5)
Implementing and integrating parts	16/03-26 (Week 7)
Testing in simulation	30/03-26 (Week 9)
Evaluate project progress and timeline	01/04-26 (week 9)
Make improvements	06/04-26 (Week 10)
Testing in simulation	20/04-26 (Week 12)
Experiments and evaluation	04/05-26 (Week 14)
Dedicated report writing	18/05-26 (Week 16)
Report correction	25/05-26 (Week 17)
Preferred submission date	29/05-26 (Week 17)
Official submission date	01/06-26 (Week 18)